DarkShield — Research summary (auto-generated)

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Abstract
We report macro-F1 0.9363662591037314 (merged confirm-shaming pass) on a stratified holdout when available (evaluated_at=2026-04-10T18:45:45.485769+00:00). Chi-square outputs quantify vertical heterogeneity in dark-pattern prevalence.

1. Introduction
DarkShield is a transparency-oriented stack that combines a labeled corpus of manipulative UI copy, a TF–IDF + random forest baseline, optional neural benchmarking, and browser-local ONNX inference. This note summarizes metrics produced by the repository's evaluation scripts.

2. Dataset
The training table ``dataset.csv`` contains approximately 5664 rows after merges from seed, scraped pages, synthetic clean lines, Princeton DAPD–inspired synthetics, FTC-themed vignettes, and CNIL-aligned examples, with cosine de-duplication and adversarial augmentation in the builder pipeline.

3. Methodology
Baseline RF macro-F1 ≈ 0.9146; DistilBERT macro-F1 ≈ 0.993. Latency (ms/snippet): RF 21.064, BERT 8.144. Conclusion recorded in JSON: RF chosen for extension deployment due to ~0x latency advantage with ~8.6% macro-F1 tradeoff vs DistilBERT on this holdout.

4. Results
Global vertical × dark-pattern association: χ^2 p-value=1.722563047493928e-53, Cramér's V=0.476275830884139. Per-pattern tests and a prevalence heatmap are saved under ``ml/models/saved/eval_results``.

5. Discussion
Patterns with significant vertical association ($p<0.01$) include: fake_urgency, confirm_shaming, hidden_costs, visual_interference, sneaking, forced_continuity, misdirection, social_proof_manipulation, bait_and_switch, roach_motel. Effect sizes should be interpreted alongside sampling bias from scrapers and synthetic rows.

6. Conclusion
The random forest baseline remains the practical deployment choice when latency and bundle size dominate; DistilBERT offers higher offline F1 at a steep per-snippet latency cost. Vertical association tests and per-pattern χ^2 probes help characterize sector-specific prevalence. See Princeton DAPD (Mathur et al., 2019), FTC (2022) dark pattern report, CNIL/EU guidance, and CPRA/CCPA rulemaking themes for regulatory context.

References
Mathur, A. et al. (2019). Dark Patterns at Scale. Princeton Web Transparency — DAPD. FTC (2022). Bringing Dark Patterns to Light. EU Digital Services Act / Omnibus consumer fairness instruments; CNIL cookie/consent guidance. California CPRA / CCPA regulations — dark pattern and deceptive design themes.

Model	Macro F1	Latency ms/snippet	Size
Random Forest	0.9146	21.064	27.19
DistilBERT	0.993	8.144	255.68